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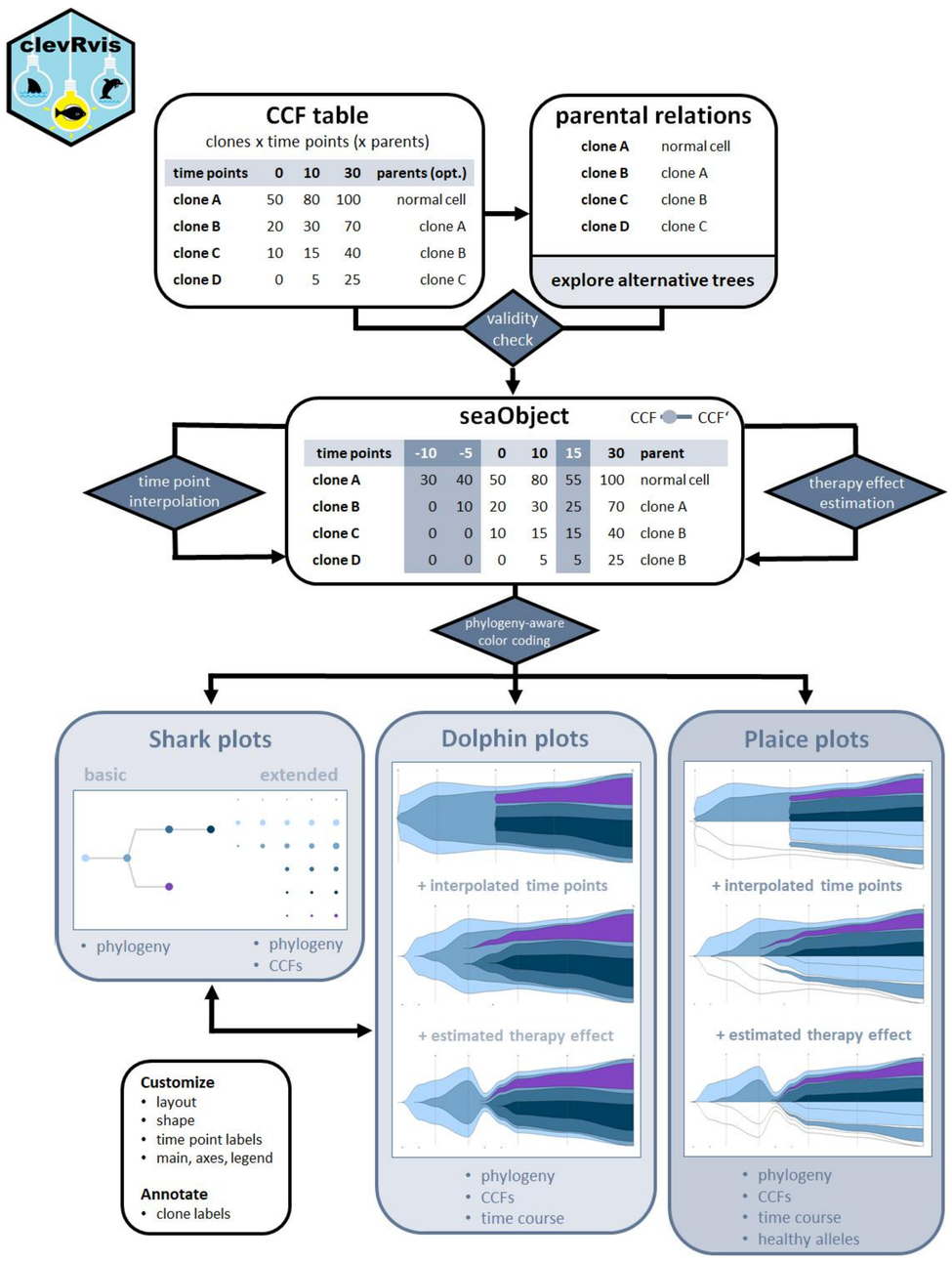
Visualizing Clonal Evolution with clevRvis

Background

Accurate reconstruction of clonal evolution is essential for the application of precision oncology. It allows for the early detection of newly developing, highly aggressive subclones that might be resistant to therapy and can potentially lead to relapse. However, analysis is characterized by challenges: Often, data on only few time points are available. As a consequence, clonal evolution is incomplete, lacking information on a tumor's development over time and its response to therapy. Furthermore, bi-allelic events, which are considered of high relevance with respect to many cancers, cannot be depicted properly.

Methods

We developed clevRvis – an R/Bioconductor package providing a wide set of visualization techniques for clonal evolution in R [1].



All analyses performed with clevRvis are based on a 'seaObject'. It contains information on the cancer cell fraction (CCF) of every clone at every time point as well as parental relations. An initial seaObject is generated on the basis of a CCF table and parental relations defined by the user. Optionally, alternative trees can be explored: taking the CCF table as a basis, all possible parental relations passing the validity check are determined. A seaObject can be extended by interpolating additional time points and/or estimating the effect of a therapy applied between two time points.

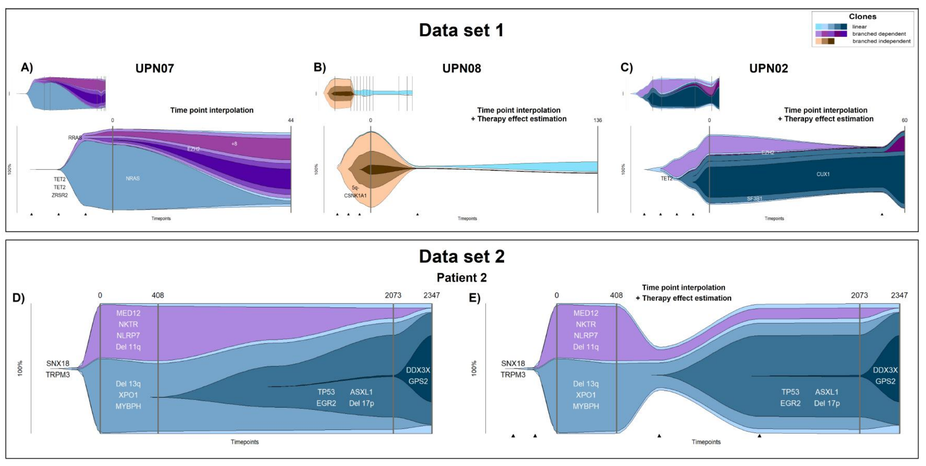
Three different representations for clonal evolution are available: shark, dolphin and plaice plots. Shark plots provide a common graph-based visualization of clonal evolution. A user may opt between the basic and the extended version (additionally showing the CCF of every clone at every time point in form of circles of different size). Dolphin plots correspond to established fish plots, showing the phylogeny, the CCFs and their joined development over time. Plaice plots provide a unique novel option for visualizing healthy vs mutated alleles with respect to clonal evolution. The dolphin plot is mirrored above and below the x axis. While the upper half represents common clonal evolution, the lower half shows the remaining healthy alleles. Optionally, shark and dolphin plots can be generated as interactively linked plots next to each other.

Results

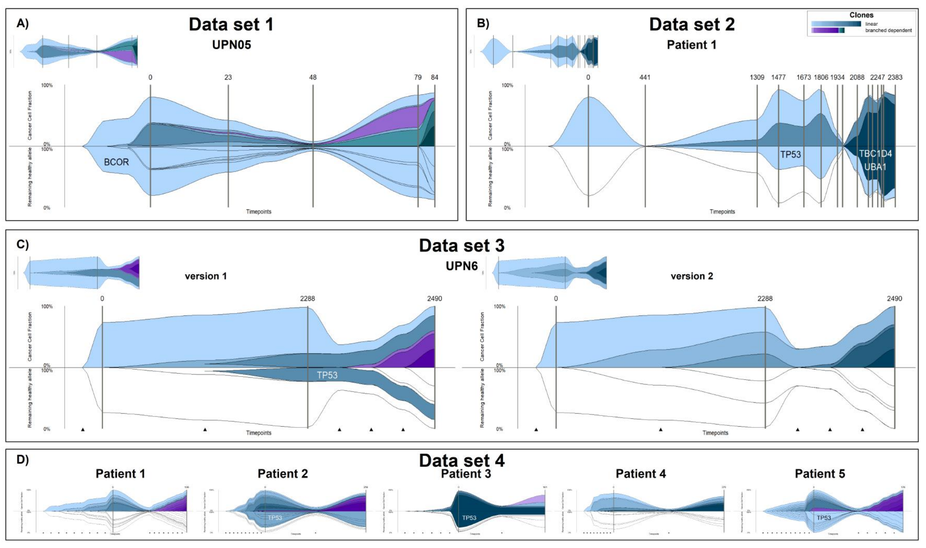
Main analysis features:

		fishplot [2]	timescape [3]	clevRvis
Visualization options	Graph		x	x
	Fish plot	x	x	x
	+ healthy alleles			x
Clonal evolution models	Linear	x	x	x
	Branched dependent	x	x	x
	Branched independent	x		x
Basic features	Graphical user interface			x
	Validity check	x		x
	Interactive plots		x	x
	Single time point		x	x
	Single clone	x		x
Advanced features	Exploring alternative trees			x
	Time point interpolation			x
	Therapy effect estimation			x
	Phylogeny-aware color coding			x

Time point interpolation and therapy effect estimation:



Detecting bi-allelic events:



Discussion

clevRvis provides an extensive set of innovative techniques for visualizing clonal evolution. To the best of our knowledge, no other comparable tool is currently available, allowing for time point interpolation, therapy effect estimation, phylogeny-aware color coding, exploring alternative trees and an allele-aware visualization of clonal evolution. In addition, clevRvis provides a shiny interface (including a detailed tutorial), allowing for user-friendly analysis of clonal evolution also by non-computer experts. Thereby, clevRvis contributes to a better understanding of a tumor's development.

References

[1] Sandmann, S. *et al.* (2023). clevRvis: visualization techniques for clonal evolution. *GigaScience*, **12**(giad020). **[2]** Miller, C.A. *et al.* (2016). Visualizing tumor evolution with the fishplot package for R. *BMC Genomics*, **17**(880). **[3]** Smith, M. *et al.* (2023). timescape: Patient Clonal Timescapes. *R package version 1.24.0*.

Availability: https://bioconductor.org/packages/clevRvis

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